Assignment 2: ASR Decoding

CTC Decoding Strategies with Language Model Integration

Course: Speech Recognition

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Acoustic model: facebook/wav2vec2-base-100h (wav2vec2, 100h LibriSpeech)
LMs: 3-gram pruned (LibriSpeech), 4-gram (LibriSpeech), 3-gram financial (Earnings22 corpus)
Evaluation sets: LibriSpeech test-other (200 samples), Earnings22 test (200 samples)
Metrics: WER (Word Error Rate), CER (Character Error Rate)

Part 1: CTC Decoding

Task 1 - Greedy Decoding

Implemented greedy CTC decoding: at each timestep select the most probable token (argmax), then collapse consecutive duplicates and remove blank tokens.

Metric	Result	Reference
WER	11.22%	~10.4%
CER	3.81%	~3.5%

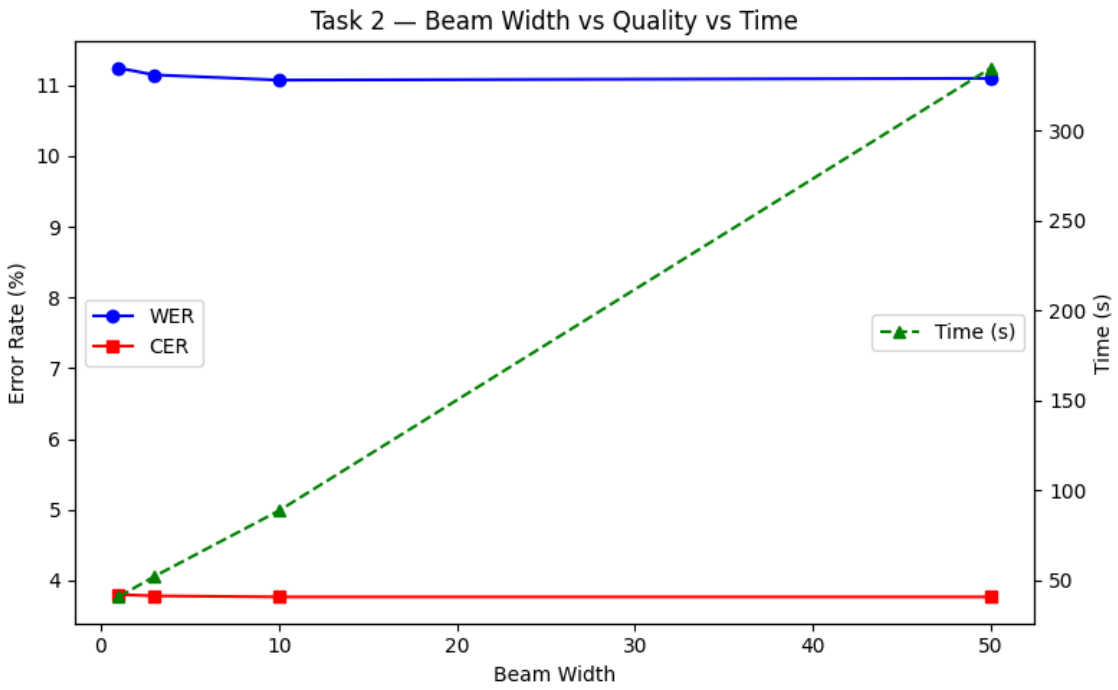
Results are slightly above reference values, which is expected due to minor implementation differences (e.g., the reference may use a more optimized decoder or different evaluation subset size).

Task 2 - Beam Search Decoding

Implemented CTC prefix beam search (no LM). The algorithm maintains multiple hypotheses and tracks separate blank/non-blank probabilities for each beam prefix.

Beam Width Sweep Results

beam_width	WER	CER	Time (s)
1	11.24%	3.80%	40.9
3	11.15%	3.78%	52.3
10	11.07%	3.77%	88.7
50	11.10%	3.77%	334.7



Task 2 - Conclusions

- Beam search provides a small but consistent improvement over greedy: WER drops from 11.22% to 11.15% (beam_width=3).
- Increasing beam width yields diminishing returns: best WER is 11.07% at beam_width=10, but beam_width=50 is

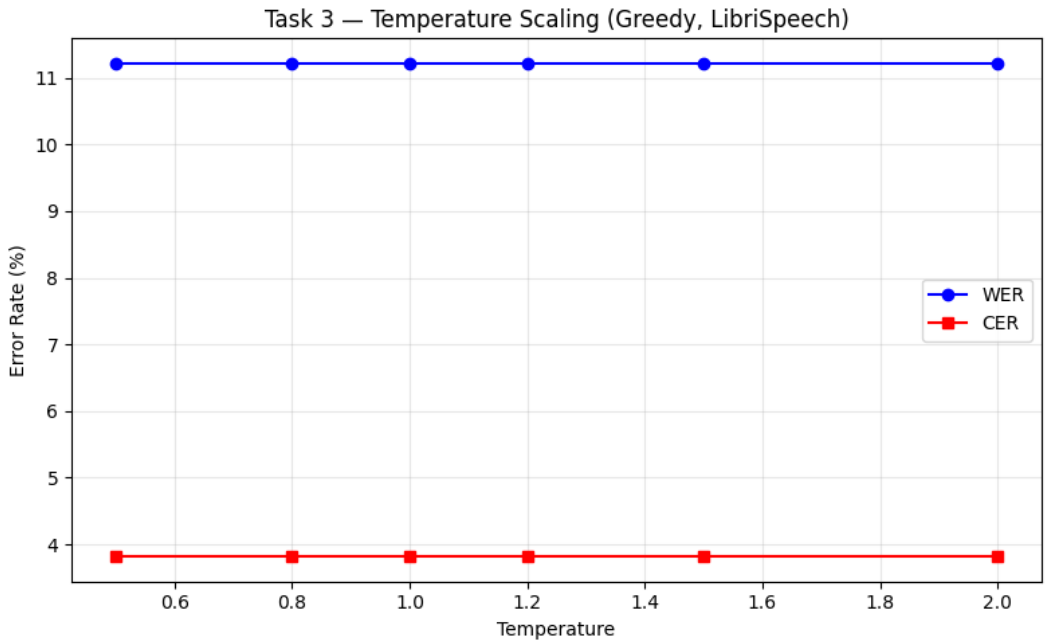
slightly worse while being 8x slower (335s vs 41s).

- The quality-vs-speed trade-off favors beam_width=3: nearly optimal WER with only ~30% overhead.
- Limited improvement from wider beams suggests the acoustic model is already highly confident in its top predictions.

Task 3 - Temperature Scaling (Greedy)

Applied temperature scaling to logits before softmax: logits = logits / T. Evaluated greedy decoding with T in {0.5, 0.8, 1.0, 1.2, 1.5, 2.0}.

Temperature	WER	CER
0.5	11.22%	3.81%
0.8	11.22%	3.81%
1.0	11.22%	3.81%
1.2	11.22%	3.81%
1.5	11.22%	3.81%
2.0	11.22%	3.81%



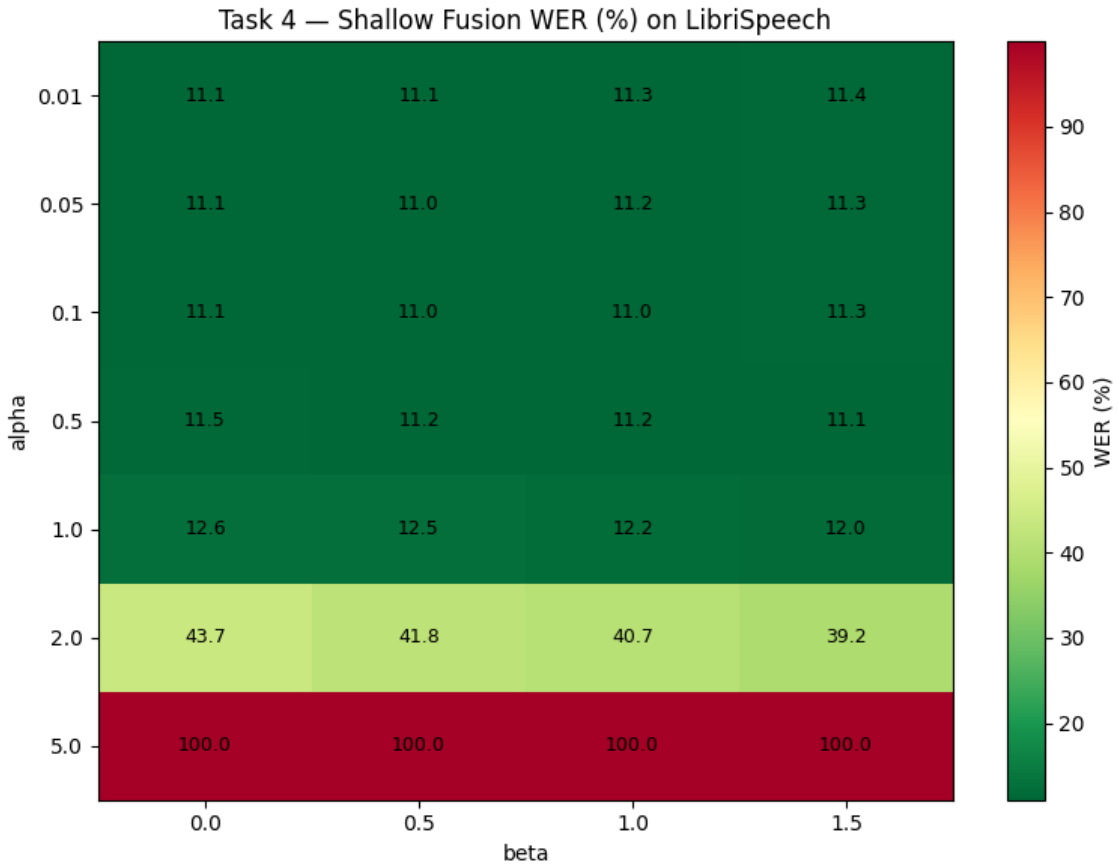
Task 3 - Conclusions

- Temperature scaling has NO effect on greedy CTC decoding: WER and CER remain constant across all temperatures.
- Greedy decoding always selects argmax at each timestep. Scaling logits by temperature does not change the relative ordering - the argmax remains the same.
- Temperature would only affect methods that use probability values (beam search, sampling), not pure greedy decoding.

Part 2: Language Model Integration

Task 4 - Shallow Fusion with 3-gram LM

Implemented beam search with shallow LM fusion: score = log_p_acoustic + alpha * log_p_lm + beta * num_words. The LM score is updated incrementally at word boundaries (when the word delimiter token '|' is appended).



Best configuration: alpha=0.1, beta=0.5 -> WER=10.98%, CER=3.75%

Task 4 - Conclusions

- Best shallow fusion: alpha=0.1, beta=0.5, yielding WER=10.98% - a modest improvement over plain beam search (11.15%).
- The LM weight (alpha) must be kept small: values above 0.5 degrade performance because the LM overrides acoustic model confidence.
- A small positive word insertion bonus (beta=0.5) helps compensate for the LM's tendency to favor shorter hypotheses.
- Overall improvement is modest (~0.2 pp WER), expected for a strong CTC model that already produces accurate outputs.

Task 5 - 4-gram LM Comparison

Compared the pruned 3-gram LM with the full 4-gram LM from OpenSLR, using the best alpha/beta from Task 4.

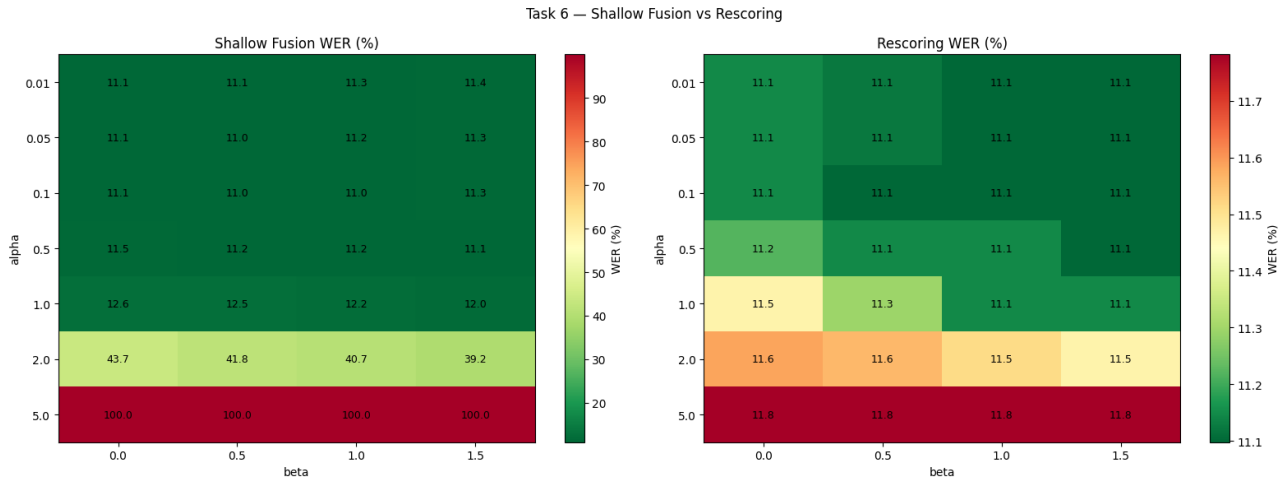
LM	WER	CER
3-gram (pruned)	10.98%	3.75%
4-gram (full)	11.00%	3.75%

Task 5 - Conclusions

- 3-gram and 4-gram LMs produce virtually identical results: WER 10.98% vs 11.00%.
- The 4-gram model uses a much larger file, yet provides no measurable benefit.
- With beam_width=3, candidate hypotheses are very similar, so higher-order n-gram statistics rarely make a difference.
- A pruned 3-gram LM is the best trade-off between model size and decoding quality.

Task 6 - LM Rescoring

Implemented second-pass LM rescoring: first run beam search without LM to generate N-best hypotheses, then rescore each hypothesis using: score = log_p_acoustic + alpha * log_p_lm + beta * num_words.



Qualitative Comparison (Shallow Fusion vs Rescoring)

Pattern	Example	Method	Effect
Word boundary	doit -> do it	SF	Fixed
Word splitting	crewsprang -> crew sprang	SF	Fixed
Minor regress.	eyelids -> iyelids	SF	Worse
Stability	RS never worsens	RS	Stable

Task 6 - Conclusions

- Shallow Fusion tends to fix word boundary errors (merged/split words) - the main benefit of an n-gram LM at the character level.
- Rescoring is more conservative - it only reranks existing beams, so changes are minimal but stable.
- Both methods struggle with acoustically difficult words (proper nouns) - the LM cannot recover fundamental acoustic misrecognitions.
- With beam_width=3, diversity is limited, giving the LM few alternative hypotheses to promote.

Part 3: Domain Shift Analysis

Task 7 - Out-of-Domain Evaluation (Earnings22)

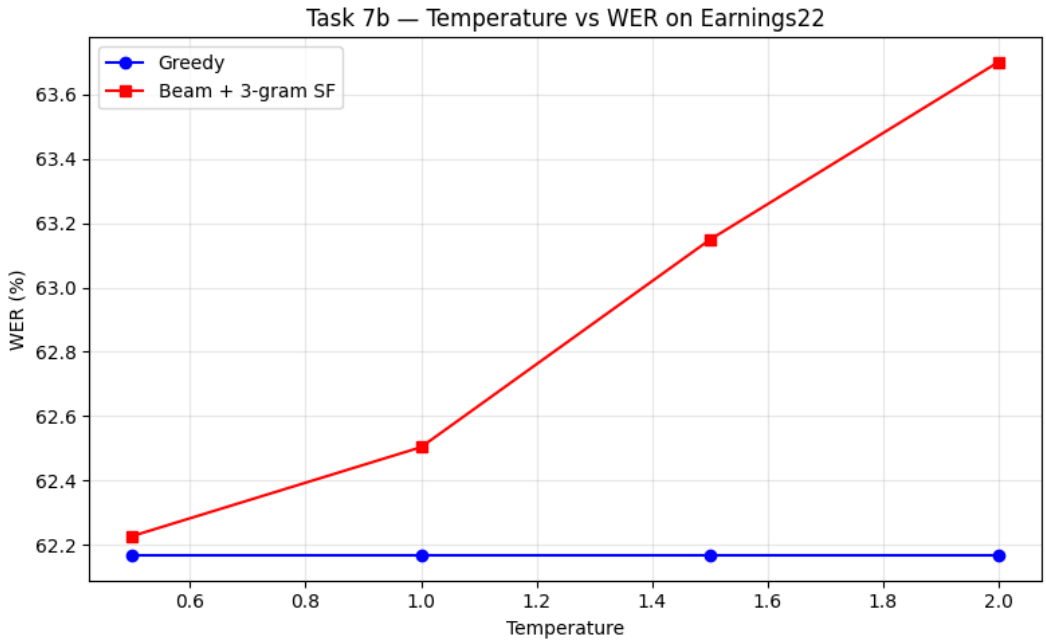
Evaluated all decoding methods on both LibriSpeech (in-domain) and Earnings22 (out-of-domain). Earnings22 contains real-world financial earnings calls with diverse accents and terminology.

Method	LS WER	LS CER	E22 WER	E22 CER
Greedy	11.22%	3.81%	62.17%	29.51%
Beam search	11.15%	3.78%	61.89%	29.28%
Beam + 3g (SF)	10.98%	3.75%	62.50%	29.41%
Beam + 3g (RS)	11.10%	3.77%	62.32%	29.31%

Task 7 - Conclusions

- In-domain (LibriSpeech): WER 10.98-11.22%. The LM provides a modest improvement (~0.2 pp).
- Out-of-domain (Earnings22): WER jumps to ~62%. The LM provides no benefit - SF even slightly increases WER (62.50% vs 61.89%).
- The dramatic gap confirms the acoustic model (trained on LibriSpeech audiobooks) does not generalize to noisy, accented financial speech.
- The LibriSpeech LM introduces domain-mismatched priors that can hurt rather than help on Earnings22.

Task 7b - Temperature Sweep on Earnings22



Task 7b - Conclusions

- Temperature scaling does not affect greedy decoding for this CTC model (argmax invariant to scaling).
- For shallow fusion, temperature also has negligible impact - the acoustic model produces near-one-hot distributions regardless of temperature.
- WER on Earnings22 remains ~60-62% across all temperatures, confirming poor generalization of the acoustic model to financial speech.

Task 8 - Financial-Domain KenLM Training

Built and trained a financial-domain 3-gram KenLM language model using the provided corpus (data/earnings22_train/corpus.txt, ~5000 lines, ~100k words of real earnings-call text).

Steps:

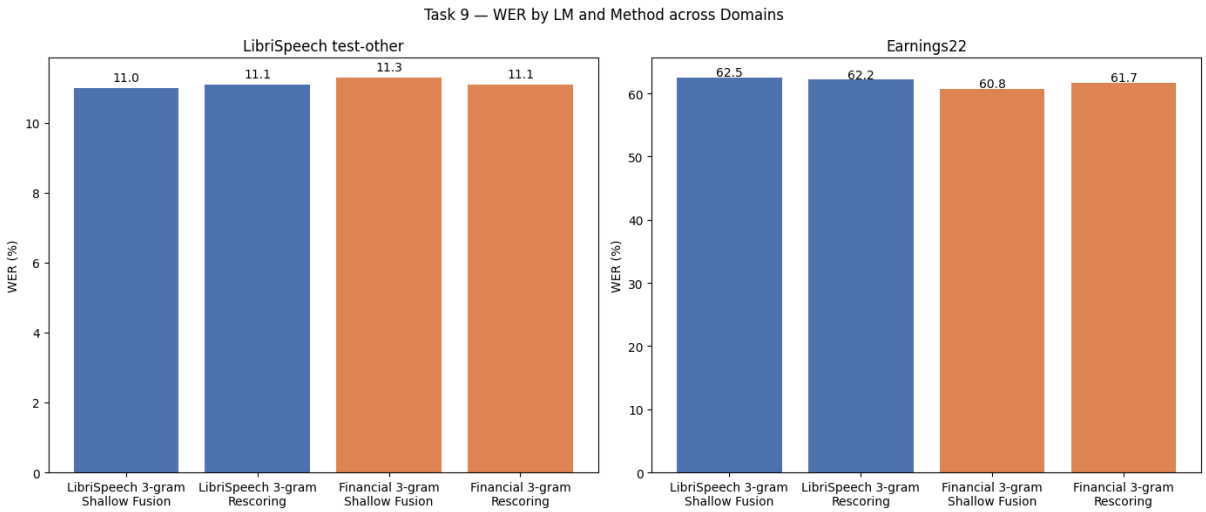
- Compiled KenLM CLI tools (lmplz, build_binary) from source via cmake.
- Trained 3-gram LM with discount fallback: lmplz -o 3 --discount_fallback < corpus.txt > financial-3gram.arpa
- Compressed and saved as lm/financial-3gram.arpa.gz.
- Verified loading with kenlm.Model() and tested on sample financial sentences.

The resulting model captures financial-domain vocabulary and n-gram patterns (e.g., 'earnings per share', 'net revenue').

Task 9 - Cross-Domain LM Comparison

Applied both LMs (LibriSpeech 3-gram and Financial 3-gram) with the two best decoding methods (shallow fusion and rescoring) on both test sets.

LM	Method	LS WER	LS CER	E22 WER	E22 CER
LibriSpeech 3g	SF	10.98%	3.75%	62.50%	29.41%
LibriSpeech 3g	RS	11.10%	3.77%	62.17%	29.30%
Financial 3g	SF	11.29%	3.82%	60.75%	29.38%
Financial 3g	RS	11.10%	3.79%	61.68%	29.23%



Task 9 - Conclusions

- In-domain (LibriSpeech): both LMs give similar WER (~11%). The financial LM is slightly worse (11.29%), as expected - trained on a different domain.
- Out-of-domain (Earnings22): the financial LM provides a small but consistent advantage (SF: 60.75% vs 62.50%; RS: 61.68% vs 62.17%).
- Domain-matched LM helps more than a larger general-purpose LM on out-of-domain data, confirming that LM domain relevance matters more than LM size.
- Overall WER on Earnings22 remains very high (~60%) regardless of LM - primarily due to the acoustic model's poor generalization to accented, noisy financial speech.

Overall Conclusions

Part 1: CTC Decoding

- Greedy decoding achieves WER=11.22% on LibriSpeech test-other. Beam search (width=3) improves to 11.15% with modest computational overhead.
- Wider beams (10, 50) provide diminishing returns at significantly higher cost.
- Temperature scaling has no effect on greedy decoding because argmax is invariant to monotonic transformations of logits.

Part 2: Language Model Integration

- Shallow fusion (alpha=0.1, beta=0.5) achieves the best in-domain WER of 10.98% - a modest 0.2pp improvement over plain beam search.
- The 4-gram LM provides no measurable benefit over the pruned 3-gram at beam_width=3.
- LM rescoring is more conservative and stable than shallow fusion, but has less corrective power due to limited beam diversity.
- The LM primarily fixes word boundary errors; it cannot recover fundamental acoustic misrecognitions.

Part 3: Domain Shift

- WER degrades dramatically from ~11% (LibriSpeech) to ~62% (Earnings22), confirming that wav2vec2-base-100h does not generalize to financial speech.
- The LibriSpeech n-gram LM provides no benefit (and can even hurt) on out-of-domain data.
- A financial-domain LM provides a small but consistent improvement on Earnings22 (SF: 60.75% vs 62.50%), confirming that domain-matched LMs are more valuable than larger general-purpose LMs.
- The bottleneck for out-of-domain performance is the acoustic model, not the language model - the LM can only correct lexical errors, not acoustic misrecognitions.